

Quasar Clustering Using Machine Learning

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Introduction

Quasars are a type of Active Galactic Nuclei (AGN) which are extremely bright centers of galaxies that are powered by supermassive black hole-consuming material. Quasars with redshifts between three and six correspond to a time when the universe was only a few billion years old. Consequently, quasar clustering could show more information about the conditions of the universe at that time and potentially the formation of the first supermassive black holes and galaxies. The pattern of quasar clustering could serve as a good indicator of the early universe’s matter distribution of galaxies, supermassive black holes, and dark matter (Osmer 1998). There is currently a discrepancy among astronomers regarding Quasar clustering at different redshifts.

Research Goal

While there is a current consensus among low-redshift Quasar clustering models, there is less consensus among those at higher redshifts based on previous research (Ross et al. 2009). Our research aims to find the possibilities of Quasar clustering between redshifts 3 and 6.

Methods

Data Cleaning

We gained our data from the 18th Data Release of the Sloan Digital Sky Survey (SDSS Collaboration 2023). We filtered our data according to the following conditions:

- $17.25 \leq r \leq 22.75$ (Carrasco et al. 2015).
- If $3 \leq \text{Redshift} \leq 4$, then $0.5 \leq g-r \leq 1.5$, $0 \leq r-i \leq 0.5$, $0.5 \leq i-z \leq 1.0$
- If $\text{Redshift} > 4$, then $0.5 \leq r-i \leq 2.0$, $1.0 \leq i-z \leq 2.5$ (Richards et al. 2006).
- $18 \leq g \leq 21$ (Hughes et. al. 2022).

We converted the celestial coordinates, RA and DEC, to galactic coordinates, latitude, and longitude. Additionally, we created a new “Distance” column using Hubble’s Law and Hubble’s Constant of 72 km/s/Mpc.

We performed a Gaussian kernel transform to identify non-linear relationships and map them as linear ones. When analyzing the distribution of quasars in each redshift range, we noticed a heavily skewed distribution towards lower redshifts. To address this, we decided to cluster our quasars using their redshifts as weights by multiplying redshift to the latitude, longitude, and distance columns. By doing this, quasars that are at higher redshifts would have heavier weights making them more significant. Lastly, we decided to standardize the Latitude, Longitude, and Distance columns. After our data cleaning process, we had 13,489 Quasars.

Machine Learning Models

We used three clustering algorithms to analyze our data: K-Means, DBSCAN, and Gaussian Mixture Model (GMM). When determining the optimal number of clusters for K-Means, we used the Elbow Method and Silhouette Score Analysis. For the DBSCAN algorithm, we used the K Nearest Neighbor Algorithm to determine the optimal eps value. Lastly, for the GMM algorithm, we calculated the Bayesian Information Criterion (BIC) to determine the optimal number of clusters.

Figures and Results

Both the Elbow Method and Silhouette Score Analysis returned six optimal clusters for the K-Means algorithm. The KNN Algorithm returned an eps value of 0.732. Using this value, DBSCAN returned one cluster and 22 outliers. Lastly, GMM returned 24 clusters.

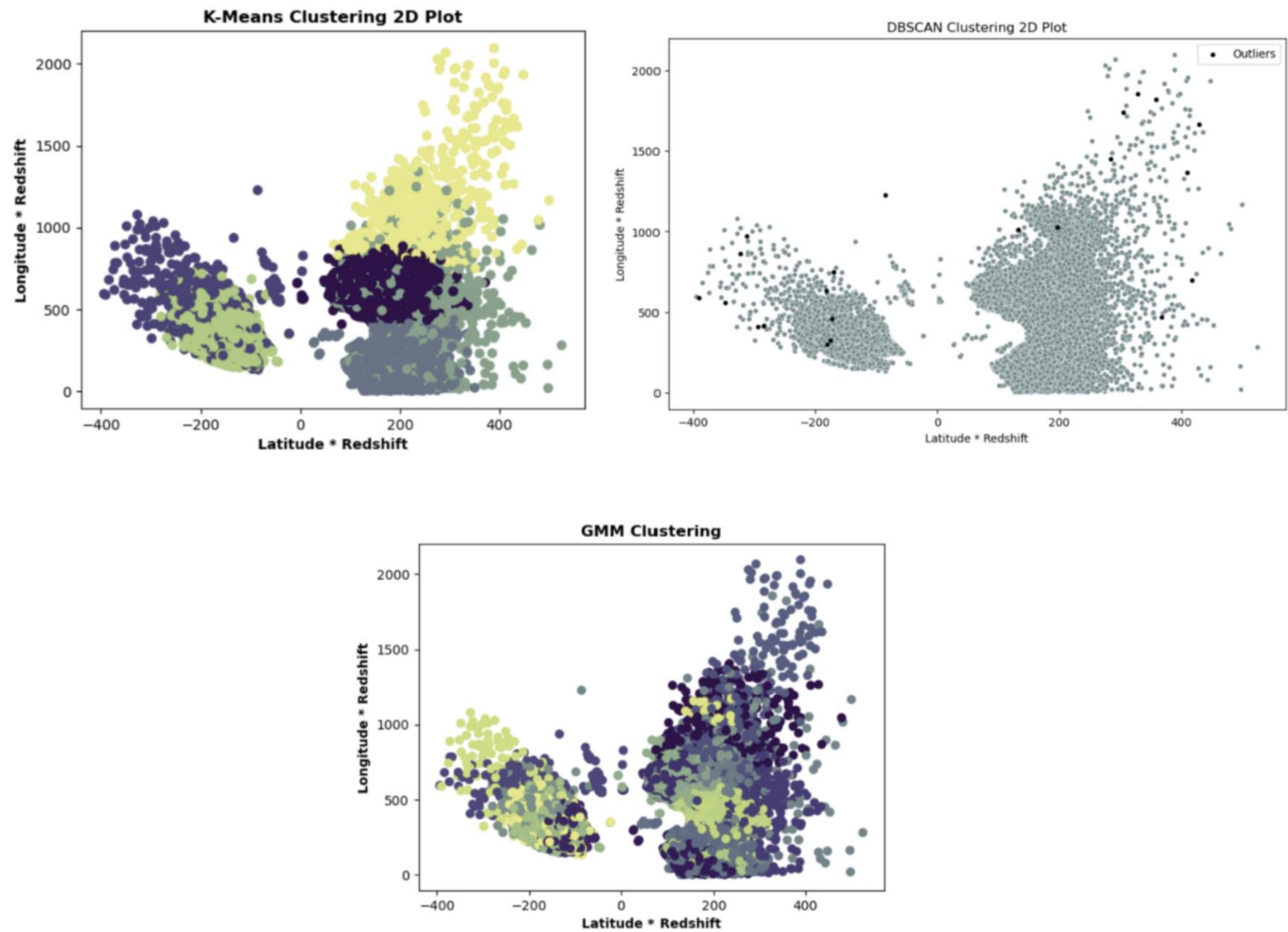


Figure 1. Visualizations of K-Means, DBSCAN, GMM Clustering Algorithms Results

When studying each algorithm’s results and respective computer analyses, we see that there’s an inconsistency among all models. Each model doesn’t return a similar number of clusters. Additionally, when we do study each cluster in the visualization plots, we see that there are no overlapping clusters (Figure 1).

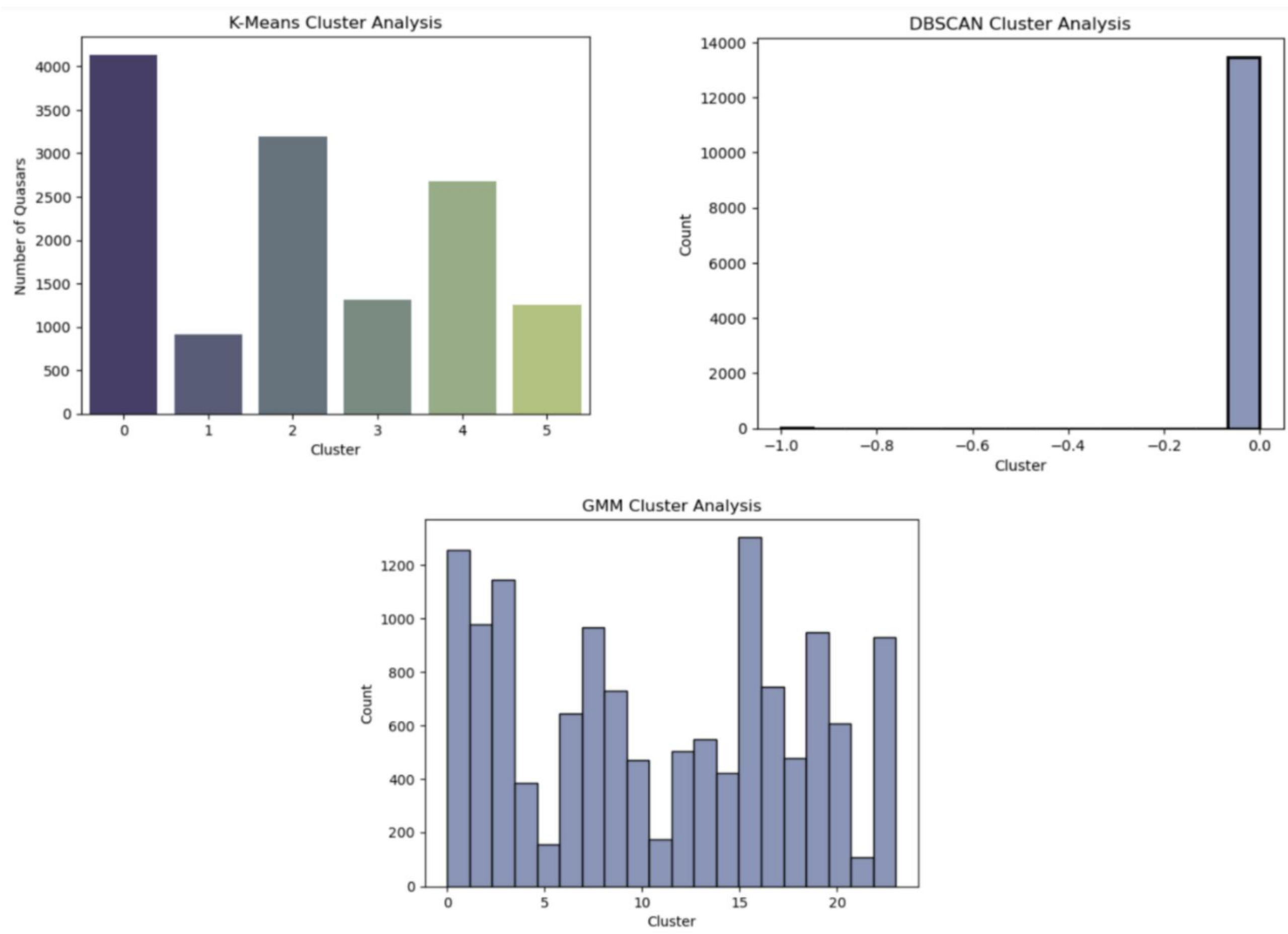


Figure 2. Algorithms Analysis Plots

Because there is a great inconsistency in the number of clusters and each cluster’s position, it is hard for us to conclude there is significant evidence of quasar clustering within this redshift range. When we also analyze the number of quasars in each cluster, the number fluctuates greatly (Figure 2).

With the K-Means and GMM algorithms, there seemed to be a relatively weak even distribution. However, the DBSCAN algorithm only returns one cluster containing most of the quasars except for a few outliers. Overall, due to strong inconsistencies, our research doesn’t point towards evidence of clustering. Some reasons for these inconsistencies could be from our dataset. For instance, when examining the 2D plots of each of our models, we see that there is a very large missing gap between latitudes -20 and 20. A main reason for this could be due to the Milky Way’s band in which the heavy dust prevented further quasars from being detected.

Conclusion

Overall, our research was able to answer our original research topic of trying to find quasar clustering systems between redshifts 3 and 6. Although our results and analyses pointed towards no significant evidence of clustering, we were still able to learn a lot more about the universe. Specifically, our results seem to indicate that the large-scale structure of the universe is more uniform than the early universe. Additionally, our results point toward future research about quasar clustering using different properties.

Our research studied quasar clustering based on spatial properties (i.e. RA and dec). However, future research could be done by studying the physical properties such as color filter magnitudes and trying to find evidence of clustering systems. From another perspective, our results could mean that there is no clustering of quasars at higher redshifts, which would contradict conclusions made by previous research. In the future, our research can be modified to study quasar clustering in smaller blocks of redshift ranges.

Acknowledgments

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